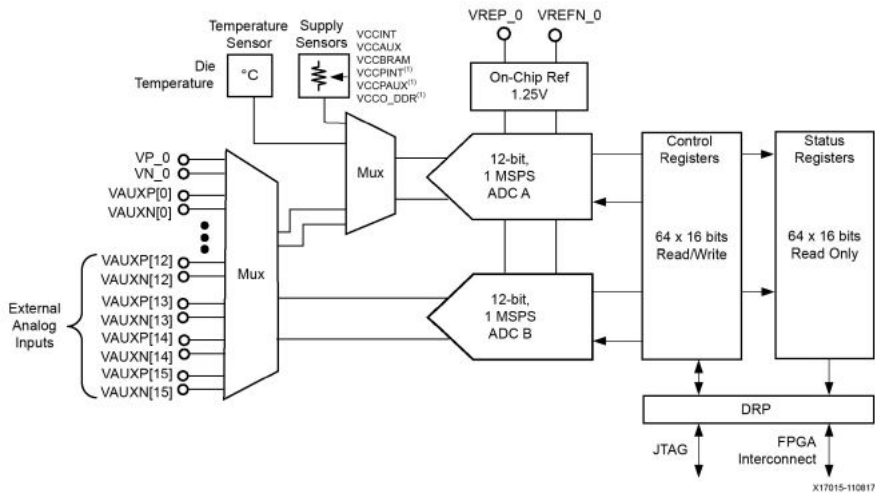


The XADC Module

(Official technical document)

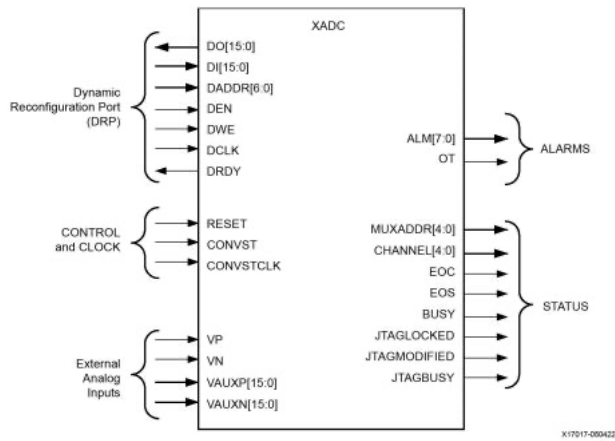
The XADC includes a dual 12-bit, 1 Mega sample per second (MSPS) ADC and on-chip sensors. The ADCs can access up to 17 external analog input channel.



Apart from a single dedicated analog input pair (VP/VN), the external analog inputs use dual-purpose I/O. These FPGA digital I/Os are individually nomin inputs when the XADC is instantiated in a design. This document refers to these analog inputs as auxiliary analog inputs. A maximum of 16 auxiliary analog available. The auxiliary analog inputs are enabled by connecting the analog inputs on the XADC primitive to the top level of the design. When enabled as ai these package balls are unavailable as digital I/Os. It is also possible to enable the auxiliary analog inputs preconfiguration (for example, for PCB diagnos JTAG TAP

V _{P_0}	Dedicated analog input	This is the positive input terminal of the <u>dedicated</u> differential analog input channel (V _P /V _N). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . This pin should be connected to GND if not used.
V _{N_0}	Dedicated analog input	This is the negative input terminal of the <u>dedicated</u> differential analog input channel (V _P /V _N). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . This pin should be connected to GND if not used.
AD0P to _AD15P_ ^{(1) (2)}	Auxiliary analog inputs/digital I/O	These are multi-function pins that <u>can support analog inputs or can be used as regular digital I/O</u> (see Figure 1-1). These pins support up to 16 positive input terminals of the differential auxiliary analog input channels (V _{AUXP} /V _{AUXN}). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . When not being used as analog inputs, these pins can be treated like any other digital I/O.
AD0N to _AD15N_ ^{(1) (2)}	Auxiliary analog inputs/digital I/O	These are multi-function pins that <u>can support analog inputs or can be used as regular digital I/O</u> (see Figure 1-1). These pins support up to 16 negative input terminals of the differential auxiliary analog input channels (V _{AUXP} /V _{AUXN}). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . When not being used as analog inputs, these pins can be treated like any other digital I/O.

XADC ports



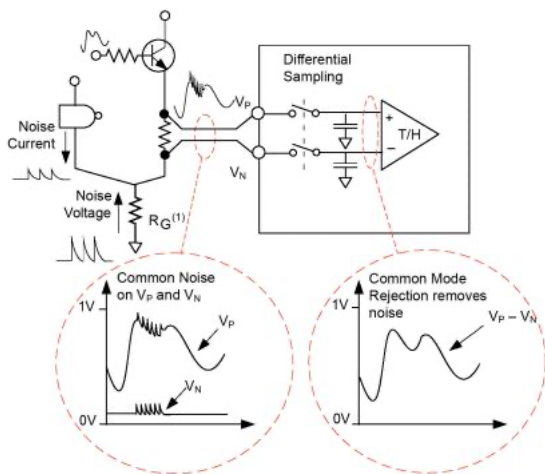
X17017-000422

The ADCs have a nominal analog input range from 0V to 1V. In unipolar mode (default), the analog inputs of the ADCs produce a full scale code of FFFh (1023) when the input is 1V. Thus, an analog input signal of 200 mV in unipolar mode produces an output code of

$$((0.2 / 1.0) \times \text{FFFh}) = 819 \text{ or } 333\text{h}$$

The ADCs always produce a 16-bit conversion result. The 12-bit data correspond to the 12 MSBs (most significant) in the 16-bit status registers. The unused 4 bits can be used to minimize quantization effects or improve resolution through averaging or filtering. The 12-bit unipolar transfer function for the ADCs. The input range to the ADCs is 0V to 1V in this mode. The ADC produces a zero code (000h) when 0V is present on the ADC input and a full scale code of all 1V is present on the input.

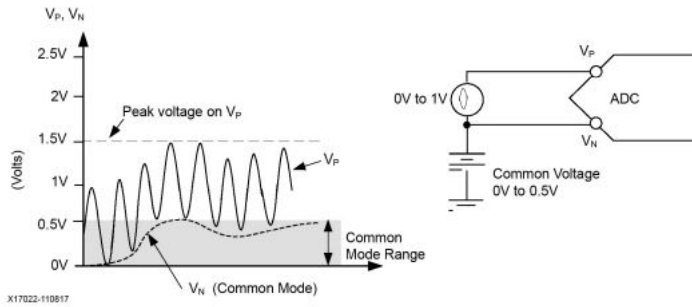
The analog inputs of the ADC use a differential sampling scheme to reduce the effects of common-mode noise signals. This common-mode rejection improves performance in noisy digital environments. To take advantage of the high common mode rejection, users need only connect VP and VN in a differential configuration.

Note 1: R_G is Common Ground Impedance.

X17020-110817

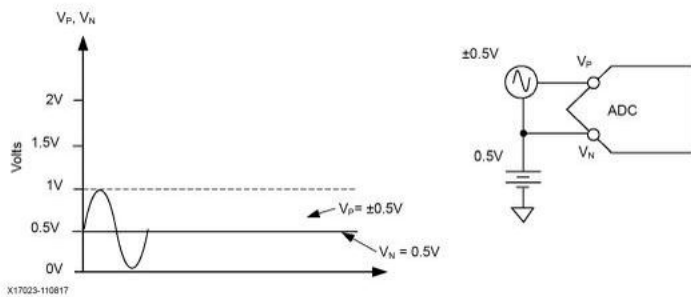
Unipolar Input Signals

When measuring unipolar analog input signals, the ADCs must operate in a unipolar input mode. This mode is selected by writing to configuration register (Registers, page 35). When unipolar operation is enabled, the differential analog inputs (V_P and V_N) have an input range of 0V to 1.0V. In this mode, the signal on V_P (measured with respect to V_N) must always be positive. Figure 2-6 shows a typical application of unipolar mode. V_N is typically connected to a local ground signal. The common mode signal on V_N can vary from 0V to +0.5V (measured with respect to GND_{ADC}). Because the differential input range is from 0V to 1V ($V_P - V_N$), the maximum signal on V_P is 1.5V.



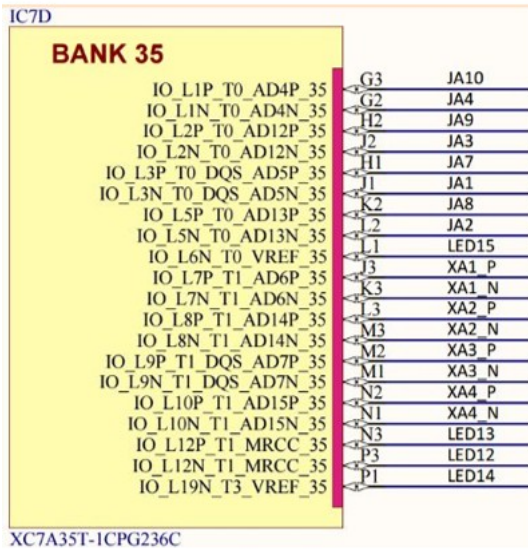
Bipolar Input Signals

The analog inputs can accommodate analog input signals that are positive and negative with respect to a common mode or reference. To accommodate the signals, the analog input must be configured to bipolar mode. Bipolar mode is selected by writing to configuration register 0 (see Control Registers). All signals must be positive with respect to analog ground (GND_{ADC}). When bipolar operation is enabled, the differential analog input ($V_P - V_N$) can have a maximum $\pm 0.5V$. The common mode or reference voltage should not exceed 0.5V in this case.



In our application, simply short GND to V_N and short OpAmp output to V_P . We are not lifting the output by adding an bias to it.

The Basys 3 board only has 4 channels of the ADC connected to header pins.



```
##Pmod Header JA
##Sch name = JA1
#set_property PACKAGE_PIN J1 [get_ports {JA[0]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[0]]}

##Sch name = JA2
#set_property PACKAGE_PIN L2 [get_ports {JA[1]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[1]]}

##Sch name = JA3
#set_property PACKAGE_PIN J2 [get_ports {JA[2]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[2]]}

##Sch name = JA4
#set_property PACKAGE_PIN G2 [get_ports {JA[3]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[3]]}

##Sch name = JA7
#set_property PACKAGE_PIN H1 [get_ports {JA[4]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[4]]}

##Sch name = JA8
#set_property PACKAGE_PIN K2 [get_ports {JA[5]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[5]]}

##Sch name = JA9
#set_property PACKAGE_PIN H2 [get_ports {JA[6]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[6]]}

##Sch name = JA10
#set_property PACKAGE_PIN G3 [get_ports {JA[7]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JA[7]]}

##Pmod Header JXADC
##Sch name = XA1_P
#set_property PACKAGE_PIN J3 [get_ports {JXADC[0]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[0]]}

##Sch name = XA2_P
#set_property PACKAGE_PIN L3 [get_ports {JXADC[1]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[1]]}

##Sch name = XA3_P
#set_property PACKAGE_PIN M2 [get_ports {JXADC[2]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[2]]}

##Sch name = XA4_P
#set_property PACKAGE_PIN N2 [get_ports {JXADC[3]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[3]]}

##Sch name = XA1_N
#set_property PACKAGE_PIN K3 [get_ports {JXADC[4]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[4]]}

##Sch name = XA2_N
#set_property PACKAGE_PIN M3 [get_ports {JXADC[5]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[5]]}

##Sch name = XA3_N
#set_property PACKAGE_PIN M1 [get_ports {JXADC[6]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[6]]}

##Sch name = XA4_N
#set_property PACKAGE_PIN N1 [get_ports {JXADC[7]]}
#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[7]]}
```

Other AD ports can be identified on the pinout of the XC7A35T - CP6236 device [here](#). More information about the pinout can be found [here](#). Typically, analog inputs are allocated evenly over banks 15 and 35. The following ports are within Bank 35.

V7	IO_L19N_T3_VREF_34	3	34	NA	NA	HR	NA
G3	IO_L1P_T0_AD4P_35	0	35	NA	NA	HR	NA
G2	IO_L1N_T0_AD4N_35	0	35	NA	NA	HR	NA
H2	IO_L2P_T0_AD12P_35	0	35	NA	NA	HR	NA
J2	IO_L2N_T0_AD12N_35	0	35	NA	NA	HR	NA
H1	IO_L3P_T0_DQS_AD5P_35	0	35	NA	NA	HR	NA
J1	IO_L3N_T0_DQS_AD5N_35	0	35	NA	NA	HR	NA
K2	IO_L5P_T0_AD13P_35	0	35	NA	NA	HR	NA
L2	IO_L5N_T0_AD13N_35	0	35	NA	NA	HR	NA
L1	IO_L6N_T0_VREF_35	0	35	NA	NA	HR	NA
J3	IO_L7P_T1_AD6P_35	1	35	NA	NA	HR	NA
K3	IO_L7N_T1_AD6N_35	1	35	NA	NA	HR	NA
L3	IO_L8P_T1_AD14P_35	1	35	NA	NA	HR	NA
M3	IO_L8N_T1_AD14N_35	1	35	NA	NA	HR	NA
M2	IO_L9P_T1_DQS_AD7P_35	1	35	NA	NA	HR	NA
M1	IO_L9N_T1_DQS_AD7N_35	1	35	NA	NA	HR	NA
N2	IO_L10P_T1_AD15P_35	1	35	NA	NA	HR	NA
N1	IO_L10N_T1_AD15N_35	1	35	NA	NA	HR	NA
N3	IO_L12P_T1_MRCC_35	1	35	NA	NA	HR	NA

It didn't show Bank 15 on this webpage but I saw a bunch of 'A' ports in Bank 14 and they are occupied by buttons and seven segment display modules on board.

However, from the Bank 35 snapshot above, channels 6 (XA1), 7 (XA3), 14 (XA2), and 15 (XA4) are available at the header pins.

We can also see corresponding information from the constraint file. The name of the ports were changed to vauxp/n x from the JXADC[] in the original c

```

207 #Pmod Header JXADC
208 #Sch name = XA1_P
209 set_property PACKAGE_PIN J3 [get_ports {vauxp6}]
210 set_property IOSTANDARD LVCMOS33 [get_ports {vauxp6}]
211 #Sch name = XA2_P
212 set_property PACKAGE_PIN L3 [get_ports {vauxp14}]
213 set_property IOSTANDARD LVCMOS33 [get_ports {vauxp14}]
214 #Sch name = XA3_P
215 set_property PACKAGE_PIN M2 [get_ports {vauxp7}]
216 set_property IOSTANDARD LVCMOS33 [get_ports {vauxp7}]
217 #Sch name = XA4_P
218 set_property PACKAGE_PIN N2 [get_ports {vauxp15}]
219 set_property IOSTANDARD LVCMOS33 [get_ports {vauxp15}]
220 #Sch name = XA1_N
221 set_property PACKAGE_PIN K3 [get_ports {vauxn6}]
222 set_property IOSTANDARD LVCMOS33 [get_ports {vauxn6}]
223 #Sch name = XA2_N
224 set_property PACKAGE_PIN M3 [get_ports {vauxn14}]
225 set_property IOSTANDARD LVCMOS33 [get_ports {vauxn14}]
226 #Sch name = XA3_N
227 set_property PACKAGE_PIN M1 [get_ports {vauxn7}]
228 set_property IOSTANDARD LVCMOS33 [get_ports {vauxn7}]
229 #Sch name = XA4_N
230 set_property PACKAGE_PIN N1 [get_ports {vauxn15}]
231 set_property IOSTANDARD LVCMOS33 [get_ports {vauxn15}]
232

```

You can see that VP_0 and VN_0 are not in the constraint file which means they might be left unused on the Basys3 board. They don't need to be in the

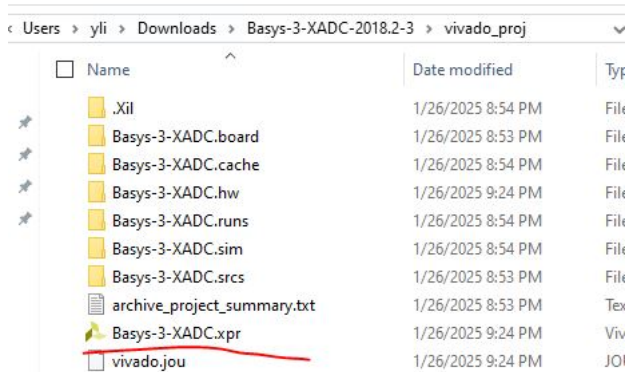
B13	VN_0	NA	0	NA	NA	CONFIG	NA
C9	VCCBATT_0	NA	0	NA	NA	CONFIG	NA
C8	TCK_0	NA	0	NA	NA	CONFIG	NA
B11	DXN_0	NA	0	NA	NA	CONFIG	NA
A13	VREFN_0	NA	0	NA	NA	CONFIG	NA
A12	VP_0	NA	0	NA	NA	CONFIG	NA
C11	CCLK_0	NA	0	NA	NA	CONFIG	NA

C. An Official Example to test

1. Load the 2018.2 version to your board for the first demo

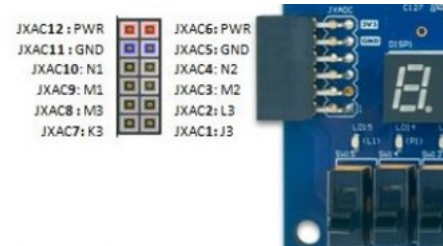
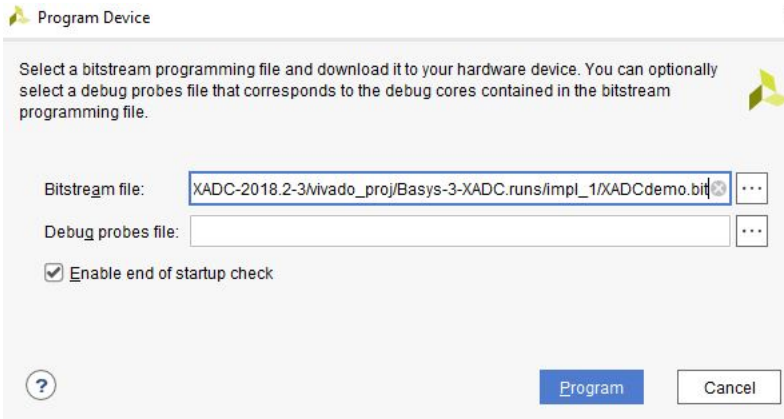
From the [official tutorial](#) I found the [2018.2 version on GitHub](#).

Download and unzip the project folder, locate the project file and double click.

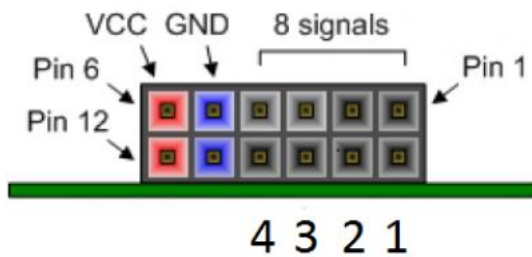


It may ask you about the compatibility issues with your 2018.3 Vivado. They may have some changes to the XADC IP. I chose to update the IP to the 2018.2 version. The window popped up.

This project folder contains the .bit file so in Vivado, directly open the hardware management to program the device using the .bit file.

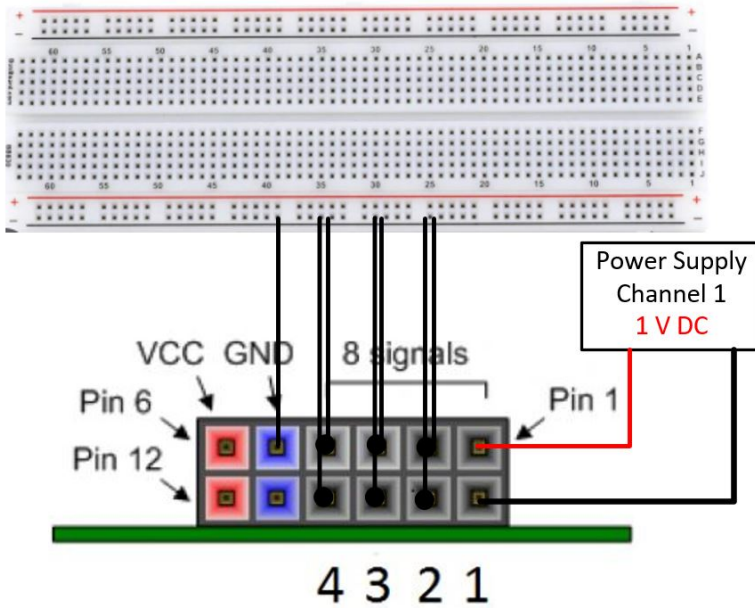


IO_L5P_T0_AD13P_35	2	JA0
IO_L5N_T0_AD13N_35	2	JA2
IO_L6N_T0_VREF_35	1	LED15
IO_L7P_T1_AD6P_35	3	XA1_P
IO_L7N_T1_AD6N_35	3	XA1_N
IO_L8P_T1_AD14P_35	3	XA2_P
IO_L8N_T1_AD14N_35	3	XA2_N



XADC has differential inputs. I used Channel 1 (or AD6, figure above) for testing purposes. All other pins must be shorted to GND to avoid crosstalk (cr). The Basys 3 GND is connected to a bread board so the unused channels can all be connected to the GND rail on the board.

A bench-top power supply output a floating 1V DC to Channel 1 (vauxp 6 and vauxn 6). The Basys 3 board should show 1.000 on the seven segment display & LED effects on the leds.



Channel Number	SW0 / SW1 Positions
XA1	Down / Down
XA2	Up / Down
XA3	Down / Up
XA4	Up / Up

Let's take a look at things in the top module of the official example.

First, from the instantiation of the xadc module inside the top module, you can tell that the 'end of conversion' (eoc_out) is shorted to 'digital enable in', which means that whenever the conversion is done, it sends a pulse to enable the next conversion. vn_in and vp_in are not available on the Basys 3 board. T and not used.

```

61 //xadc instantiation connect the
62 xadc_wiz_0 XLXI_7 (
63   .daddr_in(Address_in), //addr
64   .dclk_in(CLK100MHZ),
65   .den_in(enable),
66   .di_in(0),
67   .dwe_in(0),
68   .busy_out(),
69   .vauxp6(vauxp6),
70   .vauxn6(vauxn6),
71   .vauxp7(vauxp7),
72   .vauxn7(vauxn7),
73   .vauxp14(vauxp14),
74   .vauxn14(vauxn14),
75   .vauxp15(vauxp15),
76   .vauxn15(vauxn15),
77   .vn_in(vn_in),
78   .vp_in(vp_in),
79   .alarm_out(),
80   .do_out(data),
81   //reset_in(),
82   .eoc_out(enable),
83   .channel_out(),
84   .drv_out(readv)

```

You can tell that when 'ready' goes to 1, bits 12 - 15 are used as a counter. It is looking at the top 4 bits of the analog input to determine how many LEDs

```

83     .channel_out(),
84     .drdy_out(ready)
85 );
86
87 //led visual
88 always @(posedge(CLK100MHZ)) begin
89     if(ready == 1'b1) begin
90         case (data[15:12])
91             1: led <= 16'b11;
92             2: led <= 16'b111;
93             3: led <= 16'b1111;
94             4: led <= 16'b11111;
95             5: led <= 16'b111111;
96             6: led <= 16'b1111111;
97             7: led <= 16'b11111111;
98             8: led <= 16'b111111111;
99             9: led <= 16'b1111111111;
100            10: led <= 16'b11111111111;
101            11: led <= 16'b111111111111;
102            12: led <= 16'b1111111111111;
103            13: led <= 16'b11111111111111;
104            14: led <= 16'b111111111111111;
105            15: led <= 16'b1111111111111111;
106            default: led <= 16'b1;
107        endcase
108    end
109 end
110

```

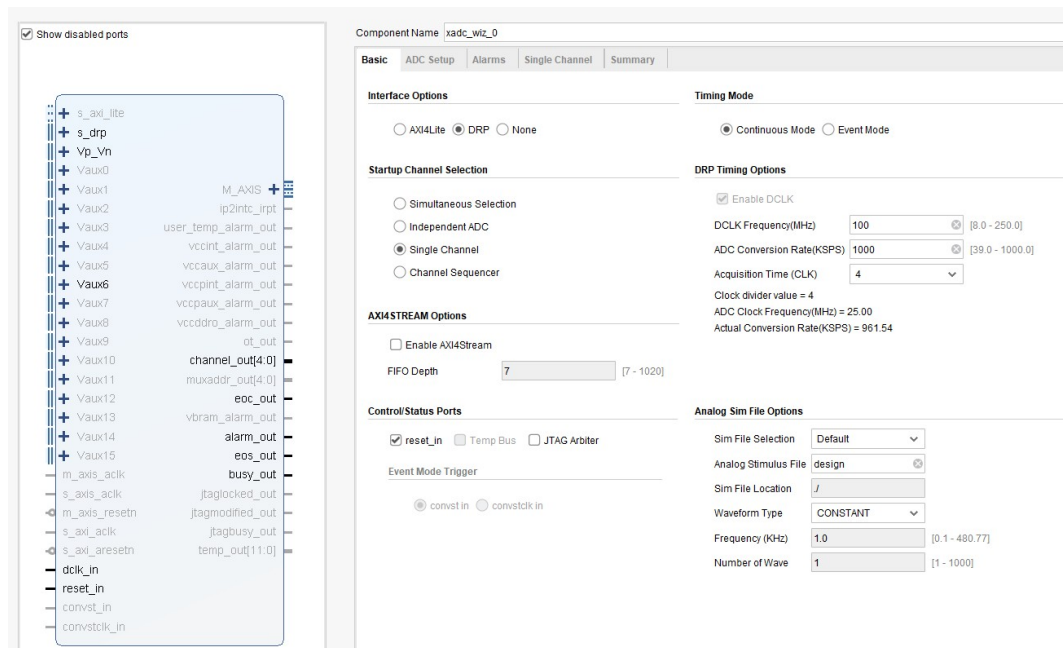
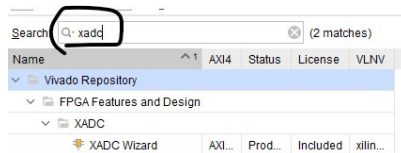
In summary, the XADC module outputs the 16-bit 'do_out' pins and connect them to the 'wire [15:0] data' vector. Only the top 12-bits are ADC data, yo the top 8 bits. This will reduce the resolution but will also simplify things.

Tasks:

Task 1: Use a potentiometer to output an analog voltage from 0 - 1 V to test the example ADC code through seven segment display (10 points) and LEDs (1 the demo video

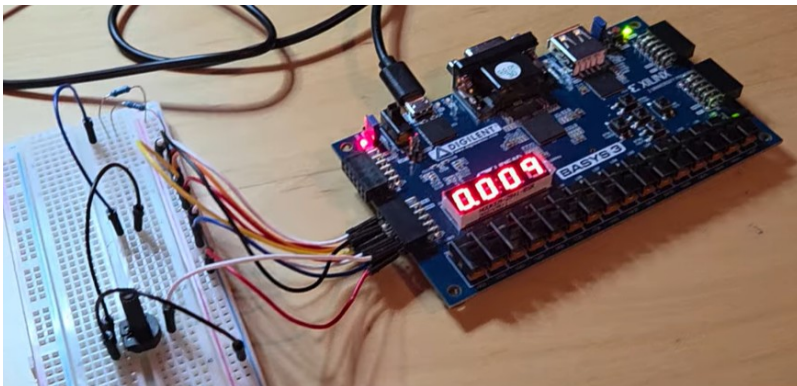
[Example code](#)

Task 2: Repeat Task 1 using the XADC IP you created and added to the project. (10 points).



) Show disabled ports

Basic		ADC Setup		Alarms		Single Channel		Summary	
Select Channel			Channel Enable			Average Enable		Bipolar	
/AUXP6 VAUXN6			☒						



Yiyan Li



Watch on